

Unsupervised Diffusion Model for Sensor-based Human Activity Recognition

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ABSTRACT

Recognizing human activities from sensor data is a vital task in various domains, but obtaining diverse and labeled sensor data remains challenging and costly. In this paper, we propose an unsupervised statistical feature-guided diffusion model for sensor-based human activity recognition. The proposed method aims to generate synthetic time-series sensor data without relying on labeled data, addressing the scarcity and annotation difficulties associated with real-world sensor data. By conditioning the diffusion model on statistical information such as mean, standard deviation, Z-score, and skewness, we generate diverse and representative synthetic sensor data. We conducted experiments on public human activity recognition datasets and compared the proposed method to conventional oversampling methods and state-of-the-art generative adversarial network methods. The experimental results demonstrate that the proposed method can improve the performance of human activity recognition and outperform existing techniques.

CCS CONCEPTS

• Computing methodologies → Unsupervised learning; • Mathematics of computing → Time series analysis.

KEYWORDS

Human activity recognition, Sensor data generation, Unsupervised learning, Statistical feature-guided diffusion model

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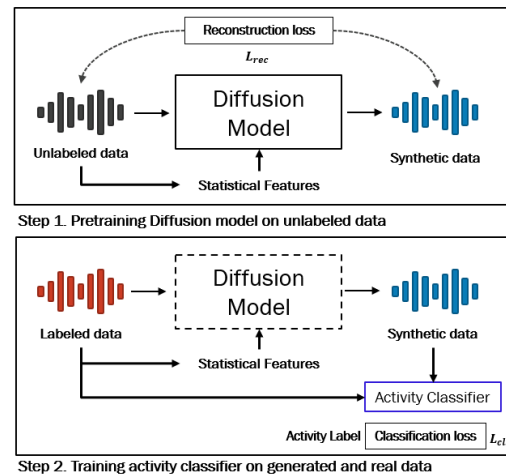


Figure 1: Overview of the statistical feature-guided diffusion model (SF-DM).

Table 1: Comparison results of human activity recognition on different datasets with 20% of real data. Results are averaged over 5 runs.

		B.L.	SMOTE	SVMS.	TimeGAN	CC-DM	SF-DM
MM-Fit	Acc.	0.826	0.845	0.840	0.827	0.835	0.849
	M.F1	0.397	0.433	0.434	0.315	0.319	0.441
PAMAP2	Acc.	0.491	0.489	0.492	0.477	0.488	0.484
	M.F1	0.430	0.352	0.419	0.412	0.383	0.435
Oppo.	Acc.	0.366	0.489	0.488	0.470	0.446	0.493
	M.F1	0.198	0.412	0.361	0.357	0.271	0.427

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